TEXORA · LECTURE SHEET

Quality Control in Apparel Manufacturing

Defect classification, in-line and end-of-line control, and acceptance sampling by ISO 2859-1: the AQL chain from lot size to accept/reject numbers, with a fully worked audit decision.

TE-3107 · APPAREL MANUFACTURING TECHNOLOGY-II · TOPIC 10 · LEVEL 3 TERM 1

1. Quality in apparel: definitions that work

Quality in a garment factory means **conformance to the specification** — measurements within tolerance, construction as approved, appearance as the sealed sample — verified at the cheapest point where a fault can still be corrected. The control system therefore has layers: **incoming** (fabric and trims), **in-line** (at the operation, by roving inspectors and operators' self-checks), **end-of-line** (100% visual and measurement check of finished garments), and **final audit** (a statistical sample of packed goods deciding the shipment). This sheet builds the statistical layer rigorously, because it is the one buyers audit.

2. Defects and their classification

CLASS	DEFINITION	APPAREL EXAMPLES
Critical	Endangers the user or breaks the law; zero tolerance — one occurrence rejects and triggers a full sort.	Needle fragment in the garment, wrong care/fibre label, mould, sharp point on children's wear.
Major	Makes the garment unsaleable or unusable as first quality; the customer would return it.	Broken zip, open seam, hole, measurement outside tolerance, wrong colour/size, large stain, missing button.
Minor	Noticeable only on close inspection; does not affect sale or use.	Small untrimmed thread, slight shading within a garment, small slub, minor pressing mark.

Inspectors work to a written **defect catalogue with photographs** and to defined viewing conditions (distance about 1 m, standard lighting, a few seconds per zone), because an unclassified judgement

cannot be audited or appealed. A garment with several defects is one **defective unit** for sampling purposes, classified by its worst defect.

3. Acceptance sampling: ISO 2859-1 and AQL

Final audits use **ISO 2859-1** single sampling for normal inspection, indexed by the **Acceptance Quality Limit (AQL)** — the worst process average, in percent defective, that is still considered acceptable as a continuing average. Apparel convention: **critical 0, major AQL 2.5, minor AQL 4.0**, at **General Inspection Level II** unless the buyer states otherwise. The procedure is a chain of three lookups: lot size + level → **code letter**; code letter → **sample size n**; n + AQL → **accept number Ac** and **reject number Re** (Re = Ac + 1). If the defective units found are ≤ Ac the lot passes; at Re it fails and is sorted/repaired and re-audited.

> **WORKED AQL EXAMPLE — VERIFY EVERY NUMBER**
>
> **Lot:** one style/colour shipment lot of **2,400 shirts**. General Inspection Level II. From ISO 2859-1 Table 1, lot band 1,201–3,200 at Level II gives **code letter K**. Table 2 (single sampling, normal): code K → **sample size n = 125**. At **AQL 2.5** (majors): **Ac = 7, Re = 8**. At AQL 4.0 (minors): Ac = 10, Re = 11. Critical defects: Ac = 0 at any sample. **Audit result:** in the 125 sampled garments the inspectors find 6 major-defective units and 12 minor-defective units, no criticals. Majors 6 ≤ 7 — pass. Minors 12 ≥ 11 — **the lot FAILS on minors** and goes for 100% re-inspection and repair of the minor class, then re-audit. One decision per defect class, each against its own AQL — that is the whole method.

LOT SIZE	CODE (LEVEL II)	SAMPLE N	AQL 2.5 AC/RE	AQL 4.0 AC/RE
501–1,200	J	80	5 / 6	7 / 8
1,201–3,200	K	125	7 / 8	10 / 11
3,201–10,000	L	200	10 / 11	14 / 15

Two cautions belong with the table. AQL sampling judges *lots as a series*: passing one audit does not certify 2.5% as a target — a factory producing at 2.4% majors will fail many audits over a season, because the plan is designed so that quality exactly at the AQL is accepted most, but not all, of the time. And the counting unit is the **defective garment**, not the number of defects: three majors on one shirt count once, under its worst class.

4. Measurement audit and in-line control

Alongside the visual sample, auditors measure a smaller set of garments at the spec's points of measurement (TE-2207 Topic 2) — any systematic out-of-tolerance reading fails the lot regardless of appearance. In-line, the factory computes **DHU (defects per hundred units)** = (defects found ÷ units inspected) × 100 at each checking point, and plots it daily; a rising DHU at one operation is the trigger for the diagnostic method of Topic 4. Common factory targets run at DHU below 5 at end-of-line after repair, with the final audit — not the in-line figure — deciding shipment.

5. The quality system around the numbers

Sampling works only inside a system: sealed approval samples and trim cards; specification and measurement charts at the line; calibrated tapes and gauges; inspector training and rotation (an inspector who checks one operation all week stops seeing it); documented repair loops (a repaired garment is re-checked 100%, not sampled); and records — audit reports, DHU charts, needle logs — retained per buyer requirements. Buyers' own auditors (and third-party inspection companies) repeat the same ISO 2859 chain independently; a factory whose internal audit matches theirs ships without drama.

Glossary

TERM	MEANING
AQL	Acceptance Quality Limit (ISO 2859-1): the worst tolerable process average, in percent defective, for a continuing series of lots.
Ac / Re	Accept number (defectives at or below which the lot passes) and reject number (at or above which it fails); Re = Ac + 1.
Code letter	The ISO 2859-1 symbol linking lot size and inspection level to a sample size.
Defective unit	A garment counted once in sampling, classified by its most serious defect.
DHU	Defects per hundred units inspected; the in-line quality rate measure.

Key points and summary

- Defects are classed critical (zero tolerance), major (AQL 2.5) and minor (AQL 4.0), judged per garment by its worst defect.
- The ISO 2859-1 chain is: lot size + Level II → code letter → sample size → Ac/Re at the chosen AQL.
- Worked standard: lot 2,400 → code K → n = 125 → majors Ac 7/Re 8, minors Ac 10/Re 11.

- AQL is a process-average concept for a series of lots, not permission to ship 2.5% defective.

Practice

Answer in your own words; brief model answers follow.

1. A lot of 5,000 garments is audited at Level II. Give the code letter, sample size, and the Ac/Re for majors at AQL 2.5 and minors at AQL 4.0, using the table in this sheet.
2. In the worked example of §3, suppose the audit had found 8 major-defective and 9 minor-defective units. State the decision and the next step.
3. An end-of-line checker inspects 400 garments and records 26 defects. Calculate the DHU.
4. Classify: (a) a needle fragment found in a child's garment; (b) a button missing; (c) a 4 mm untrimmed thread inside a seam. State the AQL or rule applied to each class.
5. Why does sampling count defective units rather than defects, and how is a garment with one major and two minor defects counted?

Answer key

1. Lot band 3,201–10,000 → code L → n = 200. Majors (AQL 2.5): Ac 10 / Re 11. Minors (AQL 4.0): Ac 14 / Re 15.
2. Majors 8 ≥ Re 8 — fail; minors 9 ≤ Ac 10 — pass. A single failing class fails the lot: the lot is held for 100% inspection/repair of majors, then re-audited.
3. DHU = 26 ÷ 400 × 100 = 6.5.
4. (a) Critical — zero tolerance; one occurrence rejects the lot and triggers a full sort and needle-control investigation. (b) Major — counted against AQL 2.5. (c) Minor — counted against AQL 4.0.
5. Sampling plans are calibrated on the proportion of defective units in the lot. Counting defects would let one bad garment fail a lot many times over. The garment counts once, as a major defective (its worst defect decides the class).